EVIDENCE AND VERIFICATION REPORT

Purpose. This report identifies the files, code locations, tests, configurations and project documents that support each significant claim in Chapters 4, 5 and 6, and states what was verified by direct inspection during this documentation pass, what was taken from recorded evidence, and what could not be verified at all.

Method. Read-only inspection. No application source file, schema, migration, configuration, environment variable or test was modified, no migration was applied, nothing was deployed, and no test suite was re-executed. The only files written are the six documents in this folder.

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1. Identity of what was inspected

Item	Value
Repository	`C:\Users\user\Desktop\Projects\student-beneficiary-system`
Branch	`fix/sbms-completion-work`
Commit	`d6b5032` (`chore(repo): drop generated logs, report PDFs and three superseded docs`)
Parent commits of interest	`f3686ec` (certificate feature), `621f23f` (state reviewed by the O10/O11 work)
Working-tree state at inspection	Clean apart from two untracked root reports (`SBMS_O10_O11_IMPLEMENTATION_REPORT.md`, `SBMS_PHASE_2_SUPERVISOR_FINDINGS_REVIEW.md`)
Other copies of the project on disk	`student-beneficiary-system-dev` (`dev`), `-admin-login-fix`, `sbms-raw-pair-model-training` — **not** used as evidence, because their documents differ
Academic sources consulted	`ICS Project II Proposal Guide 2026_V1.docx` (School template), `Project Proposal- 169613.pdf` (approved SBMS proposal, 34 pages), `CHAPTER_4[1].docx` (Chapter 4 exemplar, used for section numbering)

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2. Code evidence index

Claim	Evidence
Public registration disabled; two role-specific sign-in routes	`backend/src/routes/auth.routes.js:9-14`
bcrypt cost 12; JWT 8 hours; wrong role refused	`backend/src/controllers/auth.controller.js:26, 37-74`
Role re-read from the database; account must be active	`backend/src/middleware/auth.js:18-25`
Role gate answers 403	`backend/src/middleware/requireRole.js:1-17`
`/api/students` field-officer-only, by design and by test	`backend/src/routes/students.routes.js:8`; `backend/tests/roleAuthorization.test.js`
Registration validates eight fields, resolves policy before the transaction, takes an advisory lock, screens all candidates, returns without the token	`backend/src/controllers/students.controller.js:15-161`
Five approved schools	`backend/src/config/schools.js`; `students_school_supported_check` in `backend/src/db/schema.sql:43-45`
Screening batches at 200, validates count/range/version, raises a typed error on failure	`backend/src/services/mlDuplicateReview.js:4, 84-117`
Predictor is a child process with a 30 s timeout and fail-closed handling	`backend/src/services/duplicateCheck.js:241-301`
Aggregation `top-1` default, corroboration rules, audit counters	`backend/src/services/duplicateAggregation.js:36-144`
Priority shared by registration and queue	`backend/src/services/mlDuplicateReview.js:50-59`
Identifier masking and audited reveal	`backend/src/controllers/duplicateFlags.controller.js:37-42, 434-446`

Claim	Evidence
Terminal review decision with reason, event and audit rows	`backend/src/controllers/duplicateFlags.controller.js:452-503`
Approval gate on verification and duplicates (both sides of a pair)	`backend/src/controllers/bursaryAwards.controller.js:173-189`
Payment gate, balance ceiling, `issued` state	`backend/src/controllers/disbursements.controller.js:65-215`
Payment transition legality and discrepancy arithmetic	`backend/src/services/bursaryRules.js:1-52`
Eligibility scoring rules	`backend/src/services/eligibility.js:1-102`; `docs/eligibility-rules.md`
QR token generation, fragment URL, stripping from responses	`backend/src/controllers/students.controller.js:58-60`; `backend/src/config/publicAppUrl.js:164-177`; `backend/src/models/student.model.js:3-7`
Public verification returns only `{found}`; malformed and unknown indistinguishable	`backend/src/controllers/verification.controller.js:1-24`
Certificate snapshot, four verification states, revocation constraints	`backend/src/controllers/certificates.controller.js:53-89, 95-270`; `backend/src/models/bursaryCertificate.model.js`; migration `20261031_bursary_certificates.sql`
Separate rate-limit budgets and shared counter implementation	`backend/src/middleware/rateLimitWindow.js`; `verificationRateLimit.js`; `certificateRateLimit.js`; `loginRateLimit.js`
Security headers, CORS allowlist, body limit, single registration of `/api/verify`	`backend/src/index.js:33-67`
Sanitised error responses	`backend/src/middleware/errorHandler.js:59-78`
Readiness contract and fail-closed start-up	`backend/src/index.js:73-125`; `backend/src/services/databaseReadiness.js`; `backend/src/services/databaseStartup.js`
Schema: eight baseline tables plus migration tables, constraints and indexes	`backend/src/db/schema.sql`; fourteen files in `backend/src/db/migrations/`
Frontend routes, role-protected layout, public pages	`frontend/src/App.jsx`; `frontend/src/components/AppLayout.jsx`; `frontend/src/pages/Verification.jsx`; `frontend/src/pages/CertificateVerification.jsx`
Officer-facing screening sentence without model internals	`frontend/src/pages/StudentRegistration.jsx:7-22, 65-75`
Six-feature contract and derivation	`ml-service/feature_engineering.py:11-117`
Artefact loading, version hashing, `predict_pairs`	`ml-service/predict.py:14-98`
Container build, inference gate, no CUDA, no secrets	`Dockerfile:1-48`
Image publication gates	`.github/workflows/publish-api-image.yml`
Deployment topology (Render equivalent, documented for reference)	`render.yaml`

3. Test artefact index

Artefact	Location	Evidence it provides	Reliability note
24 Jest suites, 151 tests	`backend/tests/*.test.js` (inventory in Chapter 5 Table 5.9)	Behaviour through the real HTTP surface against PostgreSQL	Counts recorded 2026-11-01; the suites were read during this pass to confirm the described assertions exist
Database guard	`backend/tests/testDatabaseGuard.js`	Fails closed unless the target is the loopback `sbms_test` database	Read directly

Artefact	Location	Evidence it provides	Reliability note
Native test files	`backend/test/databaseReadiness.node.js`, `databaseStartup.node.test.js`, `databaseBootstrap.node.test.js`, `errorHandler.unit.test.js`, `verification.unit.test.js`, `migrate.unit.test.js`, `jwtSecret.validation.node.cjs`	43 in-process tests	Read directly; result recorded 2026-10-31
Out-of-process smoke	`backend/test/api.smoke.test.js`	3 checks against a running API	Read directly
ML service tests	`ml-service/tests/*.py` (6 files)	33 tests: features, operating point, artefact loading, training integrity, reliability	Read directly; result recorded 2026-10-31
Research helper tests	`ml-service/experiments/sbms_dataset/test_experiment.py`	7 tests	Recorded
Browser harnesses	`scripts/browser-smoke.mjs`, `qr-smoke.mjs`, `registration-fields-smoke.mjs`, `admin-reveal-smoke.mjs`	23, 9, 11 and (unrun) reveal checks	Harnesses read; results taken from project records
Training and evaluation metrics	`ml-service/evaluation/realistic_pairs_xgboost_training_metrics.json`	Hyperparameters, holdout metrics, confusion matrix, generated scenarios, limitations	Read directly; every figure in Chapter 5 §5.5.5 comes from this file
Operating-point evidence	`ml-service/evaluation/operating_point_2026-10-30.json`	Threshold table, negative-family rates, masking, blocking projections	Summarised in `docs/ml-decision-sheet-2026-10-30.md`, which was read
False-positive baseline	`ml-service/evaluation/fpr_baseline_2026-10-30.json`; `docs/ml-evaluation-baseline-2026-10.md`	0.35% of 2 000 random unrelated pairs above 0.50	Read through the decision sheet
Training integrity	`ml-service/evaluation/training_integrity_report.json`; `ml-service/check_training_integrity.py`	Group holdout, no memorisation, no single-feature label determination	Read via tests and reports
Research-only artefacts	`ml-service/model/research/rf_attachment_style_candidate.joblib`; `ml-service/experiments/sbms_dataset/artifacts/sbms_dataset_xgb_research_only.joblib`	Kept separate from production; metrics in `ML_RELIABILITY_REPORT.md` and `reports/sbms_dataset_experiment.md`	Read directly; **not** attributed to the deployed system
Production artefact	`ml-service/model/xgb_duplicate_model.joblib`	421 089 bytes, SHA-256 `6ec9f2a3…`	Presence, size and tracked status verified with `git ls-files` and a file check

4. Document evidence index

Document	Used for	Reliability assessment
`docs/FINAL_TEST_REPORT.md`	Test counts, environment, database state, outstanding tests	**Current**, but contains two generations of result; §10 supersedes §1–§3
`docs/business-workflow-verification-2026-10-30.md`	Workflow chain, the payment-clearance defect, behaviour observations	Current and specific
`docs/frontend-verification-2026-10-30.md`	23 browser checks, the priority/`risk_level` defects	Current
`docs/qr-verification-2026-10-30.md`	QR design, 10 API + 9 browser checks, the camera blocker, residual gaps	Current

Document	Used for	Reliability assessment
`docs/bursary-certificates-2026-11-01.md`	Certificate design, endpoint roles, automated coverage, unverified items	Current
`docs/azure-deployment-log-2026-10-31.md`	Deployment identity, resource configuration, live checks, the production QR defect, carry-forward items	Current; §7–§9 record unfinished operator tasks
`docs/ml-decision-sheet-2026-10-30.md`	Operating point, negative families, masking, blocking, D1–D5	Current; explicitly labels all labels synthetic
`docs/ml-evaluation-policy.md`	The five unanswered decisions and where each lands in code	Current
`docs/ml-evaluation-baseline-2026-10.md`	0.35% false-positive baseline and queue arithmetic	Current
`docs/ml-integration-and-school-identifiers.md`	School identifier problem; model interface	Current
`docs/ml_audit.md`	The supplied external dataset and the ten-column research model	Current
`ml-service/ML_RELIABILITY_REPORT.md`	Random Forest candidate metrics; the `raw_pairs.csv` label shortcut	Current, and the basis for excluding that dataset from training
`ml-service/reports/sbms_dataset_experiment.md`	Research-only experiment metrics	Current; describes a model the system does not load
`docs/SBMS_PROJECT_HANDOVER_2026-10.md`	Status matrix, defect history, phase plans	Current as a record; §12+ supersedes its earlier sections
`SBMS_O10_O11_IMPLEMENTATION_REPORT.md` (untracked)	The `/api/students` policy decision and the `/api/verify` double-registration fix	Reliable but **untracked**; gap G-28
`SBMS_PHASE_2_SUPERVISOR_FINDINGS_REVIEW.md` (untracked)	Confirmation that the SBMS-01…18 wording is unrecoverable	Reliable but **untracked**; gap G-28
`docs/verification.md`	**Superseded claims** about the predictor	Do **not** cite; gap G-08/G-23
`docs/uat-test-plan.md`	Five `PASS` results without participants	Do **not** present as acceptance evidence; gap G-24
`docs/database-recovery-rehearsal-2026-10-31.md`	Restore rehearsal procedure and result	Current
`docs/eligibility-rules.md`	Scoring factors and categories	Current; agrees with `services/eligibility.js`

5. Verification status of the chapters' major claims

Claim in the chapters	Status	How it was established
The production classifier is XGBoost over six features, artefact `6ec9f2a3…`	**VERIFIED**	`predict.py`, `feature_engineering.py`, metrics JSON, artefact size and hash
Training used 12 000 pairs with a grouped 80/20 split and an independent 12 000-pair holdout	**VERIFIED**	`realistic_pairs_xgboost_training_metrics.json`
Holdout metrics: accuracy 0.9078, precision 0.9204, recall 0.8927, F1 0.9063, PR-AUC 0.9727, ROC-AUC 0.9685, CM 5 356/463/5 537/644	**VERIFIED**	Same JSON file
The live cut-off is 0.50 and is unvalidated	**VERIFIED**	`config/settings.js:8`; `thresholds_validated: false` in the queue response; `docs/ml-evaluation-policy.md`
Registration compares against the whole registry	**VERIFIED**	`students.controller.js:69-73` (no filter); decision sheet §6
At most one flag per registration by default	**VERIFIED**	`duplicateAggregation.js:36-39, 86-117`

Claim in the chapters	Status	How it was established
A duplicate blocks approval and payment, on either side of the pair	**VERIFIED**	Controller SQL; `workflow.test.js` checks 3–5
Payments are recorded as issued, cleared separately with evidence	**VERIFIED**	`disbursements.controller.js`; `bursaryRules.js`; database `CHECK` constraints
Receipt acknowledgement is officer-assisted	**VERIFIED**	`acknowledgement_method = 'officer_assisted'` required by constraint
Student QR and certificate QR are separate namespaces, endpoints, budgets and pages	**VERIFIED**	`index.js:65-66`; controllers; two page components; two rate limiters
No beneficiary data appears in any public response	**VERIFIED**	`verification.controller.js`; `publicView()`; tests
24 suites / 151 tests passed on 2026-11-01	**RECORDED** (not re-run)	`docs/FINAL_TEST_REPORT.md` §10; `docs/bursary-certificates-2026-11-01.md` §7
43/43 native, 3/3 smoke, 33/33 and 7/7 Python	**RECORDED** (not re-run)	`docs/FINAL_TEST_REPORT.md` §2, §4
23/23, 9/9 and 11/11 browser checks	**RECORDED** (not re-run)	Frontend, QR and deployment records
The system is deployed at the stated Azure endpoints	**RECORDED**	`docs/azure-deployment-log-2026-10-31.md` §1; not independently probed in this pass
Nine defects were found and fixed during testing	**VERIFIED as records**	Each defect is documented with its root cause
The `/api/students` restriction is intentional, not a defect	**VERIFIED**	Code, frontend call sites, regression test, O10/O11 report
The Random Forest is not in production	**VERIFIED**	`predict.py` loads only `xgb_duplicate_model.joblib`
Real-world detection performance is unknown	**VERIFIED as unknown**	No adjudicated labels in the repository or either database
A phone can scan a printed QR code	**NOT VERIFIED**	No camera run recorded; gap G-11
500+ students can be screened within three seconds	**NOT VERIFIED**	No load test; gap G-12
Users find the system usable	**NOT VERIFIED**	No user study; gap G-06

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6. What was checked during this pass, and what was not

Checked directly:

1. Every code location cited in the chapters was opened and the cited behaviour confirmed.
2. The suite inventory was regenerated by counting `test(`/`it(` occurrences per file: 24 files, 151 tests, matching the recorded total.
3. Test *names* for the fourteen most relevant suites were extracted and used to write Chapter 5 Table 5.9, so the table describes assertions that exist rather than assertions that were imagined.
4. The training metrics JSON was parsed field by field; every number in Chapter 5 §5.5.5 traces to it.
5. The artefact's size, presence and tracked status were verified (`git ls-files` reports it as tracked although `.gitignore` lists the pattern — gap G-27).
6. `git status` and `git log` fixed the revision under description and recorded the two untracked reports.
7. The proposal PDF was extracted and read for objectives, scope, methodology and the four-feature Random Forest specification, so the deviation in Chapter 4 §4.1 is documented rather than assumed.

Not performed:

1. No test suite was executed, in any language.
2. No database query, migration, backup or restore was run.

3. No deployment, restart or configuration change was made.
4. No screenshot was captured and no figure was drawn.
5. No network call was made to the deployed endpoints; their status is taken from the deployment record.
6. No interview, survey or acceptance session was conducted.

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7. How to re-verify the main claims

Commands are given as the project records them; they were **not** run in this pass.

```
# 1. Fix the revision under description
git rev-parse HEAD            # expect d6b5032
git status --short            # expect only the two untracked root reports

# 2. Backend suite (needs backend/.env.test with NODE_ENV, DATABASE_URL,
#    SBMS_TEST_DATABASE_URL, and PUBLIC_APP_URL set to an https origin)
cd backend
npx jest --runInBand tests    # expect 24 suites / 151 tests

# 3. Native and smoke tests
node --test backend/test/databaseReadiness.node.js backend/test/databaseStartup.node.test.js
backend/test/databaseBootstrap.node.test.js backend/test/errorHandler.unit.test.js
backend/test/verification.unit.test.js backend/test/migrate.unit.test.js
backend/test/jwtSecret.validation.node.cjs
# expect 43/43

# 4. Frontend build
npm.cmd --prefix frontend run build     # expect exit 0

# 5. ML tests and evaluation
ml-service\.venv\Scripts\python.exe -m unittest discover -s ml-service/tests -v      # expect 33
ml-service\.venv\Scripts\python.exe -m unittest discover -s ml-service/experiments/sbms_dataset -p
test_experiment.py
cd ml-service; .\.venv\Scripts\python.exe .\evaluate_operating_point.py              # regenerates the
operating-point JSON

# 6. Artefact identity
Get-FileHash ml-service\model\xgb_duplicate_model.joblib -Algorithm SHA256
# expect 6EC9F2A3CBE6ED0DCF80C79B15F47364D9F4F7BD3E27DA25C690D0436BB0A6F9
```

Note on disambiguation: `backend/tests/` (plural) holds the Jest suites and `backend/test/` (singular, except for the Jest-named files) holds the native files. Running the native command from the repository root rather than from `backend/` changes the relative paths and will fail for that reason alone.

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8. Residual risks to the report's accuracy

Risk	Consequence	Mitigation already applied in the chapters
Recorded test counts may not reproduce on a different machine	A reader re-running the suite could see 149–150/151	Chapter 5 §5.4.3 states the instability, its mechanism and the isolation results (gap G-20)
Project records may be superseded after this pass	A citation could point at a stale statement	Every citation names a dated document, and the two known-stale documents are named as stale

Risk	Consequence	Mitigation already applied in the chapters
The deployment identity may change (resources stopped or deleted)	"Deployed" claims could become untrue	The chapters describe the deployment as demonstrated on a stated date and note the stopped-database availability model
Figures may be drawn in a way that contradicts the specifications	Visual content would misstate the system	Each figure carries an explicit content specification, including what must **not** appear
A reader may treat synthetic metrics as accuracy	Over-claiming detection capability	Every metric is labelled synthetic, and Chapter 5 §5.5.5 lists what it does not establish
Chapters 1–3 may describe features that the implementation does not have	Cross-chapter contradiction	The known conflicts are recorded here and in the gap register; Chapters 1–3 were not edited, as that is outside this task

9. Cross-chapter consistency checks performed

Check	Result
Problem in Chapter One matches the solution in Chapters Four and Five	Consistent: within-registry duplicate detection plus an evidence-bearing payment chain; the cross-constituency limitation is stated in Chapter Five §5.6.2
Objectives in Chapter One are addressed in Four, Five and Six	Consistent: each of the four objectives appears in Chapter Four §4.2, is evaluated in Chapter Five §5.5.2 and concluded in Chapter Six §6.2.3–§6.2.6
Chapter Three methodology agrees with the implementation and evaluation	Consistent with DSR plus Agile as described in the proposal §3.2–§3.3; the classifier differs (gap G-21) and the deployment architecture differs (gap G-22), both recorded
Requirements in Chapter Four agree with the implemented system	Consistent: each FR cites a route, service or table; unsatisfied items are marked in Table 4.1
Diagrams agree with the architecture and workflows	The figure specifications are derived from controller and page code; drawing them remains the author's task (gap G-03)
Implementation in Chapter Five agrees with the design in Chapter Four	Consistent: Table 5.3 maps each Chapter Four module to its implementation and test
Testing and evaluation support the conclusions in Chapter Six	Consistent: the conclusion table in `OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md` §5 cites the supporting section for each conclusion
Module, role and entity names are used consistently	Consistent: `admin`, `field_officer`, `students`, `duplicate_flags`, `duplicate_review_events`, `bursary_awards`, `bursary_award_verification_events`, `disbursements`, `disbursement_reconciliation_events`, `bursary_certificates`, `audit_logs`, `reports`

10. Summary of the evidence position

The chapters report a system whose code, schema, tests and deployment records are internally consistent and independently checkable for their implementation claims, and whose quantitative claims fall into three groups: (a) implementation facts, which were verified by reading the code; (b) test results, which are recorded with dates and commands but were not re-executed in this pass; and (c) model metrics, which were read directly from the artefact's own report and are explicitly properties of a synthetic generator. The claims that cannot be supported — real-world detection accuracy, physical QR scanning, scale performance, and user acceptance — are named as unsupported rather than inferred, and each is assigned a gap-register item with a route to closure.